Exercises

1. Suppose a sum $f+'g$ of maps $f,g:(I^n,\partial I^n)\to(X,x_0)$ is defined using a coordinate of I^n other than the first coordinate as in the usual sum $f+g$. Verify the formula $(f+g)+'(h+k)=(f+'h)+(g+'k)$, and deduce that $f+'k\simeq f+k$ so the two sums agree on $\pi_n(X,x_0)$, and also that $g+'h\simeq h+g$ so the addition is abelian.

2. Show that if $\varphi:X\to Y$ is a homotopy equivalence, then the induced homomorphisms $\varphi_*:\pi_n(X,x_0)\to\pi_n(Y,\varphi(x_0))$ are isomorphisms for all n. [The case $n=1$ is Proposition 1.18.]

3. For an H–space (X,x_0) with multiplication $\mu:X\times X\to X$, show that the group operation in $\pi_n(X,x_0)$ can also be defined by the rule $(f+g)(x)=\mu(f(x),g(x))$.

4. Let $p:\widetilde{X}\to X$ be the universal cover of a path-connected space X. Show that under the isomorphism $\pi_n(X)\approx\pi_n(\widetilde{X})$, which holds for $n\geq 2$, the action of $\pi_1(X)$ on $\pi_n(X)$ corresponds to the action of $\pi_1(X)$ on $\pi_n(\widetilde{X})$ induced by the action of $\pi_1(X)$ on $\widetilde{X}$ as deck transformations. More precisely, prove a formula like $\gamma p_*(\alpha)=p_*(\beta_{\tilde{\gamma}}(\gamma_*(\alpha)))$ where $\gamma\in\pi_1(X,x_0)$, $\alpha\in\pi_n(\widetilde{X},\tilde{x}_0)$, and γ_* denotes the homomorphism induced by the action of γ on $\widetilde{X}$.

5. For a pair (X,A) of path-connected spaces, show that $\pi_1(X,A,x_0)$ can be identified in a natural way with the set of cosets αH of the subgroup $H\subset\pi_1(X,x_0)$ represented by loops in A at x_0.

6. If $p:(\widetilde{X},\widetilde{A},\tilde{x}_0)\to(X,A,x_0)$ is a covering space with $\widetilde{A}=p^{-1}(A)$, show that the map $p_*:\pi_n(\widetilde{X},\widetilde{A},\tilde{x}_0)\to\pi_n(X,A,x_0)$ is an isomorphism for all $n>1$.

7. Extend the results proved near the beginning of this section for the change-of-basepoint maps β_γ to the case of relative homotopy groups.

8. Show the sequence $\pi_1(X,x_0)\longrightarrow\pi_1(X,A,x_0)\xrightarrow{\partial}\pi_0(A,x_0)\longrightarrow\pi_0(X,x_0)$ is exact.

9. Suppose we define $\pi_0(X,A,x_0)$ to be the quotient set $\pi_0(X,x_0)/\pi_0(A,x_0)$, so that the long exact sequence of homotopy groups for the pair (X,A) extends to $\cdots\to\pi_0(X,x_0)\to\pi_0(X,A,x_0)\to 0$.

(a) Show that with this extension, the five-lemma holds for the map of long exact sequences induced by a map $(X,A,x_0)\to(Y,B,y_0)$, in the following form: One of the maps between the two sequences is a bijection if the four surrounding maps are bijections for all choices of x_0.

(b) Show that the long exact sequence of a triple (X,A,B,x_0) can be extended only to the term $\pi_0(A,B,x_0)$ in general, and that the five-lemma holds for this extension.

10. Show the 'quasi-circle' described in Exercise 7 in §1.3 has trivial homotopy groups but is not contractible, hence does not have the homotopy type of a CW complex.

11. Show that a CW complex is contractible if it is the union of an increasing sequence of subcomplexes $X_1\subset X_2\subset\cdots$ such that each inclusion $X_i\hookrightarrow X_{i+1}$ is nullhomotopic, a condition sometimes expressed by saying X_i is contractible in X_{i+1}. An example is

S^∞, or more generally the infinite suspension $S^\infty X$ of any CW complex X, the union of the iterated suspensions $S^n X$.

12. Show that an n-connected, n-dimensional CW complex is contractible.

13. Use the extension lemma to show that a CW complex retracts onto any contractible subcomplex.

14. Use cellular approximation to show that the n-skeletons of homotopy equivalent CW complexes without cells of dimension $n+1$ are also homotopy equivalent.

15. Show that every map $f : S^n \to S^n$ is homotopic to a multiple of the identity map by the following steps.

(a) Use Lemma 4.10 (or simplicial approximation, Theorem 2C.1) to reduce to the case that there exists a point $q \in S^n$ with $f^{-1}(q) = \{p_1, \cdots, p_k\}$ and f is an invertible linear map near each p_i.

(b) For f as in (a), consider the composition gf where $g : S^n \to S^n$ collapses the complement of a small ball about q to the basepoint. Use this to reduce (a) further to the case $k = 1$.

(c) Finish the argument by showing that an invertible $n \times n$ matrix can be joined by a path of such matrices to either the identity matrix or the matrix of a reflection. (Use Gaussian elimination, for example.)

16. Show that a map $f : X \to Y$ between connected CW complexes factors as a composition $X \to Z_n \to Y$ where the first map induces isomorphisms on π_i for $i \le n$ and the second map induces isomorphisms on π_i for $i \ge n+1$.

17. Show that if X and Y are CW complexes with X m-connected and Y n-connected, then $(X \times Y, X \vee Y)$ is $(m+n+1)$-connected, as is the smash product $X \wedge Y$.

18. Give an example of a weak homotopy equivalence $X \to Y$ for which there does not exist a weak homotopy equivalence $Y \to X$.

19. Consider the equivalence relation $\simeq_w$ generated by weak homotopy equivalence: $X \simeq_w Y$ if there are spaces $X = X_1, X_2, \cdots, X_n = Y$ with weak homotopy equivalences $X_i \to X_{i+1}$ or $X_i \leftarrow X_{i+1}$ for each i. Show that $X \simeq_w Y$ iff X and Y have a common CW approximation.

20. Show that $[X, Y]$ is finite if X is a finite connected CW complex and $\pi_i(Y)$ is finite for $i \le \dim X$.

21. For this problem it is convenient to use the notations X^n for the n^{th} stage in a Postnikov tower for X and X_m for an $(m-1)$-connected covering of X, where X is a connected CW complex. Show that $(X^n)_m \simeq (X_m)^n$, so the notation X_m^n is unambiguous. Thus $\pi_i(X_m^n) \approx \pi_i(X)$ for $m \le i \le n$ and all other homotopy groups of X_m^n are zero.

$$\begin{tikzcd} X_m \arrow[r] \arrow[d] & X_m^n \arrow[d] \\ X \arrow[r] & X^n \end{tikzcd}$$

22. Show that a path-connected space X is homotopy equivalent to a CW complex with countably many cells iff $\pi_n(X)$ is countable for all n. [Use the results on simplicial approximations to maps and spaces in §2.C.]

23. If $f: X \to Y$ is a map with X and Y homotopy equivalent to CW complexes, show that the pair (M_f, X) is homotopy equivalent to a CW pair, where M_f is the mapping cylinder. Deduce that the mapping cone C_f has the homotopy type of a CW complex.

4.2 Elementary Methods of Calculation

We have not yet computed any nonzero homotopy groups $\pi_n(X)$ with $n \geq 2$. In Chapter 1 the two main tools we used for computing fundamental groups were van Kampen's theorem and covering spaces. In the present section we will study the higher-dimensional analogs of these: the excision theorem for homotopy groups, and fiber bundles. Both of these are quite a bit weaker than their fundamental group analogs, in that they do not directly compute homotopy groups but only give relations between the homotopy groups of different spaces. Their applicability is thus more limited, but suffices for a number of interesting calculations, such as $\pi_n(S^n)$ and more generally the Hurewicz theorem relating the first nonzero homotopy and homology groups of a space. Another noteworthy application is the Freudenthal suspension theorem, which leads to stable homotopy groups and in fact the whole subject of stable homotopy theory.

Excision for Homotopy Groups

What makes homotopy groups so much harder to compute than homology groups is the failure of the excision property. However, there is a certain dimension range, depending on connectivities, in which excision does hold for homotopy groups:

Theorem 4.23. *Let X be a CW complex decomposed as the union of subcomplexes A and B with nonempty connected intersection $C = A \cap B$. If (A,C) is m-connected and (B,C) is n-connected, $m,n \geq 0$, then the map $\pi_i(A,C) \to \pi_i(X,B)$ induced by inclusion is an isomorphism for $i < m+n$ and a surjection for $i = m+n$.*

This yields the **Freudenthal suspension theorem**:

Corollary 4.24. *The suspension map $\pi_i(S^n) \to \pi_{i+1}(S^{n+1})$ is an isomorphism for $i < 2n-1$ and a surjection for $i = 2n-1$. More generally this holds for the suspension $\pi_i(X) \to \pi_{i+1}(SX)$ whenever X is an $(n-1)$-connected CW complex.*

Proof: Decompose the suspension SX as the union of two cones C_+X and C_-X intersecting in a copy of X. The suspension map is the same as the map

$$\pi_i(X) \approx \pi_{i+1}(C_+X, X) \longrightarrow \pi_{i+1}(SX, C_-X) \approx \pi_{i+1}(SX)$$